

Dubai Land Department API & Data Opportunity Audit

HLabs DXB registry mirror, outbound call volume, field coverage, missed data, reliability findings, and a brokerage/proptech product roadmap

System audited: `hlabs-dxb`

Audit date: 28 July 2026

Primary external host: `gateway.dubailand.gov.ae`

Evidence: source trace, database state, sync history/logs, targeted live HTTP probes, and official DLD open-data/API pages

Revision: project open-data integration added and validated on 28 July 2026

Bottom line. The code now uses three DLD endpoint paths. The new projects snapshot is efficient - four POST calls produce a union of 815 unique current-window projects - while an ordinary office/broker registry sync is still hundreds of calls and a deep sync is more than ten thousand. Project data materially expands the product from brokerage prospecting into supply, developer, launch and completion intelligence.

Executive summary

3

unique DLD REST resource paths currently integrated

9

implemented request shapes; 7 are used and 2 point lookups are dormant

756

combined daily logical calls: 752 registry + 4 projects

10,248

combined deep-day logical calls, before retries

Critical reliability finding. The scheduled run started 28 July 2026 at 03:34:58 UTC, ran for 9,939,228 ms (about 2h45m39s), made 50 HTTP attempts, and failed on DNS resolution. Before failing, it fetched 10,168 unique offices and inserted 25, yet the failed `SyncRun` row records `officesSeen=0`. Because office writes are committed before the broker crawl finishes, the database can represent a partial snapshot.

The successful run on 26 July made **4,545 actual HTTP attempts**. Its configuration reconciled 3,000 offices; the logical minimum was about 3,048 calls (12 office pages + 36 broker pages + 3,000 office-roster lookups). The remaining approximately 1,497 attempts were retries, a roughly 49% overhead. Current code limits regular reconciliation to 1,200, but only 704 offices met the selection rule at audit time, so the present regular-run estimate is 752 logical calls, with a hard ceiling near 1,248 before retries.

The current database contains 10,196 office records, 33,977 broker records and **815 project records**. The project snapshot completed successfully in 53.7 seconds with exactly four external attempts and no retries. Its four source result sets contained 359 start-date matches, 321 end-date matches, 203 adoption-date matches and 217 completion-date matches (1,100 rows before deduplication).

Commercial conclusion

The combined registry and project data is unusually valuable for B2B prospecting and supply intelligence. The best near-term products are a **new-brokerage radar**, **developer/project launch monitor**, **completion-risk and handover watchlist**, **broker-movement/hiring intelligence**, **licence-expiry workflow**, and **brokerage campaign planner by area/project**. Transactions, rents, units/buildings, developer licensing, valuations, listing validation and rental-index data remain the highest-value joins.

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1. Scope, evidence and counting definitions

What was reviewed

- `lib/dld/client.ts` : host, authentication parameter, headers, pagination, retries, endpoint wrappers and concurrency.
- `lib/dld/projects-client.ts` and `lib/dld/projects-sync.ts` : project POST contract, four-date union, retries, normalization, provenance and sync telemetry.
- `lib/dld/sync.ts` : full crawl, per-office reconcile selection, write order, tracking and request metrics.
- `lib/dld/normalize.ts`, `lib/dld/derive.ts`, and `prisma/schema.prisma` : raw-to-database mapping, normalization, derived facts and lead scoring.
- Installed scheduled tasks, the two recorded registry runs, and `logs/sync-daily.log`.
- Small read-only live calls to confirm office, broker and project request/response contracts and create saved Postman examples.
- Official DLD Real Estate Data and API Gateway pages for adjacent data opportunities.

How “number of API calls” is defined

Measure	Meaning	Audited answer
Unique endpoint paths	Distinct URL resources on the DLD gateway.	3
Implemented request shapes	Different parameter combinations exposed by client helpers.	9
Runtime-used request shapes	Three registry shapes plus four required project date-selector shapes.	7
Logical calls	One intended page/lookup, excluding retries.	~756 combined daily; ~10,248 on a deep day
Actual HTTP attempts	Every outbound fetch attempt, including retries. This is what <code>requestCount</code> records.	4,545 in the only successful recorded run

Important limitation. No new full office/broker sync was launched during the audit because it would create thousands of external requests and mutate the registry mirror. The project sync was launched because its complete current-window snapshot is only four calls; it wrote 815 unique records.

2. API inventory and runtime flow

Request	Path / parameters	Runtime?	Purpose and behavior
Full office crawl	<code>/offices/</code> <code>pageSize</code> , <code>pageIndex</code>	Used	Pages the licensed-office directory and deduplicates by <code>RealEstateNumber</code> .

Full broker crawl	<code>/brokers/ pageSize , pageIndex</code>	Used	Pages broker cards and deduplicates by <code>CardNumber</code> .
Office roster	<code>/brokers/ officeNumber , pageSize=500</code>	Used	Re-pulls one office's roster because the bulk endpoint is not lossless.
Single office	<code>/offices/ officeNumber , pageSize=5</code>	Dormant	<code>fetchOffice()</code> exists, but no project code imports it.
Single broker	<code>/brokers/ cardNumber , pageSize=5</code>	Dormant	<code>fetchBrokerByCard()</code> exists. It verifies the returned card because an unknown number may fall back to an unfiltered list.
Projects by date selector (4 variants)	<code>POST /open- data/projects P_DATE_TYPE=1..4 , full-year dates, P_TAKE=-1</code>	Used	Fetches start, end, adoption and completion matches; unions by <code>PROJECT_NUMBER</code> and stores which selectors matched.

Common request contract

- Host: `https://gateway.dubailand.gov.ae` .
- Credential-like parameter: `consumer-id` . The current value is embedded in source as a fallback and documented as public.
- Browser-like `User-Agent` is required; plain clients can receive HTTP 403.
- Paging: `pageIndex` is zero-based. Other common cursor names are silently ignored.
- Default timeout is 90 seconds; maximum five attempts; linear backoff of 1.5s, 3s, 4.5s and 6s.
- Roster lookups execute with concurrency 12.
- Projects use `consumer-id` as a header and a JSON POST body. Dates must be `MM/DD/YYYY` ; an empty or placeholder body fails.
- The project export contract accepts `P_TAKE=-1` , returning the selected date window in one response. The four calls run sequentially to avoid load-related timeouts.

Runtime sequence

1. Crawl and immediately upsert all office pages.
2. Crawl and upsert all broker pages; create office stubs for office numbers observed only on broker rows.
3. Select offices for roster reconciliation and call the broker endpoint once per selected office.
4. Deactivate records absent from a successful full snapshot.
5. Recompute rollups, lead scores and daily facts; update broker-card high-water mark.
6. Independently, call the project endpoint four times for the source year, union/deduplicate, hash changes, upsert all 22 DLD business fields plus raw source and write a `ProjectSyncRun` .

3. External call volume

Bulk crawl calculation using live totals

Stage	Live reported rows	Page size	Data pages	Terminal/empty page	Logical calls
Offices	10,178	1,000	11 (indexes 0-10)	1	12
Brokers	34,093	1,000	35 (indexes 0-34)	1	36
Bulk crawl total					48

The extra terminal calls occur because the gateway returns slightly short/repeated pages, so the deduplicated count may never equal `TotalRowCount`. The crawler intentionally does not treat a short page as completion.

Regular run at the audited snapshot

The regular selection rule includes active offices licensed within 180 days, offices with zero active brokers, and offices first observed in the current run, capped at 1,200. Existing data produced **704 eligible offices**. The largest office currently has 468 active brokers, below the roster page size of 500, so each current roster is one logical request.

Component	Logical requests
Bulk office + broker crawl	48
Per-office roster reconciliation	704
Current estimate, before retry overhead	752
Configured ceiling if the 1,200 cap is reached	1,248

Project snapshot calculation

Date selector	DLD code	Rows returned	Logical calls
Start date	1	359	1
End date	2	321	1
Adoption / registration date	3	203	1
Completion date	4	217	1
Union	-	1,100 returned / 815 unique	4

The 815-project union contains 560 active, 122 finished, 122 pending and 11 other-status projects across 65 areas and 452 developers. DLD requires a date type and date window, so missing from this union does not mean

a historical project ceased to exist; the sync intentionally does not deactivate project rows.

Deep run and schedule

The deep run reconciles every active office. With 10,196 active office rows, its current logical minimum is **10,244 requests** (48 bulk + 10,196 roster lookups).

Scheduled workload	Frequency	Logical requests before retries
Projects open-data sync	Daily at 07:00	4
Regular registry sync	Daily at 07:15	~752 at snapshot; max ~1,248
Deep registry sync	Sunday at 04:30	~10,244
Weekly total	7 project + 7 regular + 1 deep	~15,536 current; ~19,008 at regular cap
Average month	365/12 day equivalent	~67,522 current; ~82,622 at regular cap

Observed attempts versus logical calls

Recorded run	Status	Logical baseline	Actual attempts	Observation
26 Jul 2026 manual	Success	~3,048 (3,000 roster targets)	4,545	~1,497 extra attempts, approximately 49% overhead.
28 Jul 2026 scheduled	Failed	Did not finish bulk broker crawl	50	DNS <code>ENOTFOUND</code> ; long timeout/retry sequence.

A simple scaling of the successful-run ratio would put the present 752-call regular run near 1,120 attempts and the deep run near 15,275 attempts. This is only a sensitivity estimate: retry rates are load- and concurrency-dependent and may worsen on the deep run.

4. Data extracted today

Project data retained

Category	Fields retained	Commercial value
Identity and sponsor	Project number; English/Arabic project name; developer number and bilingual name; master project.	Canonical project/developer CRM, portfolio rollups and matching.
Dates	Start, planned end, adoption/registration, inspection and completion dates.	Launch calendar, registration cohorts, inspection recency and handover watchlists.

Financial and compliance	Declared project value, escrow account number, project status and project type.	Developer/project segmentation, escrow visibility and campaign prioritization.
Delivery	Percent completed, bilingual description and recorded completion.	Progress monitoring, delay/risk hypotheses and buyer communication.
Location and supply	Bilingual area/zone authority; counts of lands, buildings, villas, units and total inventory.	Area supply pipeline, large-project ranking and brokerage capacity planning.
Lineage	Raw source row, source year, source hash, date-selector matches, first/last seen and sync run telemetry.	Auditable freshness, change detection and safe interpretation of partial date coverage.

Publication terminology: DLD does not return a separate web `publishedAt` field. The application labels `ADOPTION_DATE` as “Published / adopted” with an explicit note that it is the project registration/adoption date, not proof of webpage publication time.

Office/brokerage data retained

Category	Fields retained or derived	Commercial value
Identity	RERA real-estate number, licence number, English and Arabic names.	Deduplication, bilingual search and account identity.
Licence lifecycle	Original issue date, expiry date, active/deactivated state, first/last seen.	New-firm radar, renewal outreach and churn/dissolution monitoring.
Contactability	Office phone/fax, contact person names, contact mobile/email.	Outbound sequencing and owner/contact routing.
Digital presence	Website, logo, YouTube, Facebook, Instagram, Twitter/X, LinkedIn, WhatsApp.	Digital maturity and service-gap scoring.
Quality/status	Office rank ID/name, awards count, licensed activities.	Segmentation, reputation signals and product eligibility.
Derived intelligence	Broker counts, 30/90-day first-card momentum, earliest/latest current card dates, lead score/tier, field-level changes.	Prioritization, growth signals and CRM automation.

Broker data retained

Category	Fields retained or derived	Commercial value
Identity	Card number, English/Arabic name, photo.	Licensed-person identity and CRM verification.
Licence lifecycle	Current card issue/expiry, first/last seen, active/deactivated state, high-water-based <code>isNewCard</code> .	New-agent prospecting and renewal workflow.

Contactability	Phone, mobile and email.	Direct broker outreach.
Office linkage	Office RERA/licence number, names, issue/expiry, office rank.	Roster, firm-size context and broker-movement detection.
Broker quality	Card rank ID/name and awards count.	Influencer/top-producer segmentation.
Derived intelligence	Lead score/tier and field-change history.	Prioritized broker queues and movement alerts.

Normalization applied

- Naive gateway dates are anchored to UTC to preserve the printed Dubai calendar day in downstream SQL buckets.
- Placeholder dates outside 1990-2100 become null.
- Phones are normalized toward +971 . . . ; known placeholder numbers are discarded.
- Email addresses are lowercased and basic-format validated; invalid values become null.
- URLs without a scheme are promoted to HTTPS when they contain a dot; non-URL values become null.
- Hyphen/zero/null-like strings are collapsed to null.

Observed coverage in the current mirror

Office field	Present	Coverage	Broker field	Present	Coverage
Contact email	10,161 / 10,196	99.7%	Email	33,972 / 33,977	>99.9%
Contact mobile	2,158 / 10,196	21.2%	Mobile	33,966 / 33,977	>99.9%
Office phone	2,286 / 10,196	22.4%	Phone	20,788 / 33,977	61.2%
Website	727 / 10,196	7.1%	Photo	33,977 / 33,977	100%
Instagram	285 / 10,196	2.8%	Card rank	33,977 / 33,977	100%
WhatsApp	628 / 10,196	6.2%	Office linkage	33,977 / 33,977	100%
Logo	10,185 / 10,196	99.9%	Issue date	33,975 / 33,977	99.99%
Activities	10,185 / 10,196	99.9%	Expiry date	33,977 / 33,977	100%

Interpretation. The low office website/social/WhatsApp coverage creates a high-value sales signal, but it must first be separated into “DLD has no value” versus “the business truly has no presence.” Enrichment and validation are required before pitching a digital gap as fact.

5. Returned data not retained

Fields visible in the same live responses

Field	Where	Current handling	Recommendation
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OfficeLogo	Broker row	Declared in TypeScript but discarded by broker normalization.	Retain only if broker-level exports need the office brand without joining to Office ; otherwise document intentional de-duplication.
SearchMatchTypeNameAr/En	Broker row	Not declared in interface and discarded.	Add to raw schema-drift tests; store only if a supported search mode makes it non-null and commercially meaningful.
RowIndex	Office and broker rows	Not stored.	Normally safe to omit; capture in raw landing logs when diagnosing missing/duplicate pages.
Row-level TotalRowCount	Office and broker rows	Not stored.	Use for consistency checks against the envelope; no need in the core entity table.
TimeStamp	Envelope	Parsed by TypeScript type but not persisted.	Persist as sourceTimestamp per request/run for freshness and lineage.
Errors	Envelope	Declared but never validated.	Fail or quarantine a page when non-empty even if HTTP status is 200.
ExceptionId	Envelope	Ignored.	Persist in request diagnostics and alerts.
ActionUserId	Live envelope	Not declared.	Add to tolerant raw schema monitoring; likely operational metadata, not an entity field.
Token	Envelope	Ignored.	Continue not storing unless DLD formally documents it; never log if it becomes sensitive.
Envelope TotalRowCount	Envelope	Used for crawl control/logging, not persisted structurally.	Store per-run reported totals and unique counts to quantify coverage drift.
timeStamp , responseCode , validationErrorsList	Project envelope	Validated in memory but not retained per request.	Persist per-request manifests if contractual lineage or forensic replay is required.
additionalData	Project envelope	Currently null and not retained.	Schema-monitor it; store only if DLD begins returning meaningful content.

Data lost through cleaning

The normalized database intentionally loses raw formatting and invalid/placeholder values. That is appropriate for operations, but it makes later data-quality analysis harder. A compact raw landing layer (JSON payload, request URL without secrets, source timestamp, page index, response checksum and schema version) would preserve evidence while the normalized tables remain clean.

6. Additional DLD datasets and APIs worth pursuing

Official public open-data families

DLD's Real Estate Data page publishes views for transactions, rents, projects, valuations, land, buildings, units, brokers and developers. h1abs-dxb now ingests the project family through the same JSON endpoint used by the official frontend, in addition to the broker/office directory. The other families remain dataset opportunities and need their own contract, rate and compliance review.

Dataset	Available fields highlighted by DLD	High-value product outcome
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Transactions	Transaction number/date/type/subtype, ready/off-plan, freehold, usage, area, property type/subtype, amount, transaction/property size, rooms, parking, nearby landmarks, buyer/seller counts, master project/project.	Comparable-sales engine, price/sq.m., liquidity/velocity, off-plan vs secondary demand, area/project momentum, valuation support.
Rents	Registration/start/end dates, new/renew version, area, contract/annual amount, freehold, size/type/subtype, rooms, usage, landmarks, parking, units, project.	Rent comps, gross-yield models, renewal behavior, tenant-demand heatmaps and landlord/broker rent guidance.
Projects	Project/developer IDs and names, dates, project type/value, escrow account, status, completion %, inspection/completion dates, description, area, authority, land/building/villa/unit totals, master project.	Integrated now: supply pipeline, launch calendar, completion risk, developer intelligence and broker campaign planning.
Valuations	Total property value, area/size, procedure year/number, date, amount, transaction size, type/subtype.	Valuation-to-sale spread, model calibration and lender/investor analytics.
Land	Land type/subtype/size, ready/off-plan, pre-registration number, freehold, zip, master project, project number/name, area, zone.	Development-site intelligence, zoning/product segmentation and land comp analytics.
Buildings	Size/common/built-up area, levels, shops/flats/offices, pools/elevators, creation date, registration/freehold/leasehold, floors/rooms/parking, land/project/area/zone.	Building inventory graph, amenity scoring, brokerage specialization and property-management targeting.
Units	Unit/building/land/project IDs, subtype/size/balcony/common area, floor/room/parking, creation date, registration/freehold/leasehold, municipality, area/zone.	Canonical unit catalog, deduplication, listing QA, inventory search and pricing features.
Broker open data	Broker number/name, gender, licence start/end, website, phone/fax, office number/name.	Gender and historical licence fields not in the current gateway model; useful for completeness cross-checks.
Developer	Developer number/name, registration date, licence source/type/number/dates, website/phone/fax, chamber number.	Developer CRM, compliance/renewal alerts and project sponsor due diligence.

Official subscription API opportunities

API family	DLD-described capability	Recommended use	Access note
Dubai Brokers	Two endpoints for real-time broker-card and office enquiry/validation.	Replace reliance on an undocumented public gateway with a supported contract and SLA.	Active DLD business account; AED 30,000 + VAT/year shown.
Rental Index	Current/historical rental prices and market indicators across property types/regions.	Rent calculator, landlord renewal advice, broker pricing tools and yield models.	Subscription/pre-requisites apply.
Trakheesi	Listing Validation API and delisted-property records by timeframe.	Permit validation, compliant publishing, stale/fake listing detection and delisting alerts.	Trade-licence requirements; AED 30,000 + VAT/year shown.
Ejari	Issue, renew and terminate tenancy contracts plus print/cancel/lookups.	Brokerage/property-management transaction workflow, not merely analytics.	Requires eligible activities, local office and PMC/landlord association.

Mollak/Oqood	Service-charge, receipts/invoices/community/owner sync; developer/bank workflows.	Property-management, community-finance and developer products.	Restricted to eligible entities; not a general lead-data source.
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Do not mix access classes. Public downloadable data, the currently used public gateway, and paid DLD subscription APIs have different terms, prerequisites and reliability expectations. Each new source needs its own licence/terms review, rate plan, data-retention policy and customer-facing attribution.

7. Brokerage and proptech use cases

Prioritized products from data already collected

Rank	Product / insight	How it is derived	Who benefits / monetization
1	New Brokerage Radar	Original office issue date + first seen + contactability + activities + digital gaps.	Internal sales, vendors and new firms; subscription feed or qualified-lead product.
2	Roster & Broker Movement Intelligence	Daily roster snapshots and changes in <code>RealEstateNumber</code> , with first/last seen.	Brokerages recruiting agents, CRMs and competitor-intelligence teams.
3	Licence Expiry Workbench	Office/card expiry windows, active state, contact channels and change history.	Brokerage compliance teams and brokers; workflow/SaaS seats.
4	Digital Readiness Score	Website/social/WhatsApp availability, contactability, office size and age.	Web/CRM/marketing service sales and brokerage self-assessment.
5	Brokerage Growth & Hiring Signal	Exact-from-observation new cards, roster deltas and office-size bands.	Proptech vendors, recruiters and competitor strategy teams.
6	Verified Broker Directory / API	Card, photo, rank, office association and freshness checks.	Consumers, portals and compliance teams; API usage pricing.
7	White-space Territory Planner	Activities, rank, firm size, licence age and digital gaps; add area data later.	Brokerage expansion and partner/channel teams.
8	Account Change Alerts	Contact, rank, website, WhatsApp, expiry and office-move diffs.	CRM users; premium alerting/automation tier.

Products unlocked by additional data

- **Area/project market cockpit:** price/sq.m., rent/sq.m., sales volume, transaction velocity, off-plan share, yields and project completion risk.
- **Broker performance context:** rank brokers and offices against the markets/projects they specialize in, using permitted non-personal aggregations.
- **Compliant listing operations:** permit validation, listing-to-unit matching, delisting/staleness alerts and duplicate/fraud detection.
- **Developer and launch intelligence:** project pipeline, completion %, inspection dates and developer licence/portfolio monitoring.
- **Investor analytics:** comparable transactions, rent yield, price/rent indexes, liquidity and pipeline supply.

- **Property-management stack:** Ejari lifecycle, Mollak service-charge workflows and owner/community operations for eligible customers.

Immediate products from the new project data

- **Developer launch radar:** group newly adopted projects by developer, area, project type, value and unit count; route relevant launches to brokerage clients.
- **Completion and handover watchlist:** compare planned end, recorded completion and percent completed to flag likely handover windows or schedule slippage for due diligence.
- **Brokerage campaign planner:** identify areas with concentrated upcoming unit supply and help brokerages allocate agents, content and lead-generation budgets.
- **Developer portfolio scorecards:** portfolio size, declared value, status mix, completion distribution, inspection recency and escrow coverage.
- **Inventory opportunity ranking:** prioritize high-unit projects, villa-heavy schemes, active low-completion projects or projects approaching completion.
- **Source-grounded project pages/API:** searchable detail, bilingual descriptions, schedule, progress, area/authority, inventory and downloadable CSV for brokerage analysts.

Conclusions the present data can support

- **Who is newly licensed?** Strong for offices across historical issue dates; strong for newly observed broker cards only after the high-water baseline began.
- **Who is reachable?** Strong for broker email/mobile and office email; weaker for office mobile/phone.
- **Who appears digitally underserved?** Useful lead hypothesis, not proof, until external web/social enrichment validates absence.
- **Which firms are hiring or losing brokers?** Strong after repeated complete roster observations; weak if a run is partial or reconciliation skips an office.
- **Who may need renewal/compliance help?** Strong from current expiry dates, subject to freshness.
- **Which brokerages are highest-value prospects?** Reasonable heuristic from age, size, reachability, digital gap and momentum; it needs outcome feedback to become predictive.
- **Which current-window projects create supply and campaign opportunities?** Strong within the explicit four-selector source-year coverage, with project/developer, schedule, progress, area, value and inventory available.

Conclusions the present data cannot support responsibly

- True historical first-licence date for brokers whose cards renewed before monitoring began.
- Broker/office sales performance, transaction volume, commission, inventory or area specialization.
- Transaction-derived property values, achieved sales velocity, rent yields, brokerage market share or buyer demand. Declared project value and construction progress are available, but they are not sales performance.
- A complete historical universe of every Dubai project. The endpoint requires a date selector and source-year window; the union is current-window coverage.
- Proof that a missing DLD website/social field means the business has no such presence.
- A fully consistent “as of now” registry count while partial failed runs can commit data.

8. Technical, data-quality and compliance findings

Priority	Finding	Impact	Recommended control
P0	Run is not snapshot-atomic.	Office data can be refreshed while broker data remains stale; customer claims and derived signals can be inconsistent.	Land raw pages into run-scoped staging tables; validate completeness; atomically promote a successful snapshot.
P0	Latest scheduled run failed after 2h45m.	Freshness breach and long resource occupancy; daily task returned a non-zero result.	DNS/network preflight, circuit breaker, shorter bounded timeout for DNS failures, alerting, and resume-from-page checkpoints.
P0	Failed-run metrics discard partial progress.	<code>officesSeen=0</code> conflicts with log evidence of 10,168 fetched and 25 inserted.	Persist stage counters incrementally or write partial result counters in the catch path.
P1	Envelope logical errors are ignored.	HTTP 200 can be accepted even if <code>Errors / ExceptionId</code> signals an application failure.	Validate envelope before using <code>Response</code> ; quarantine and alert on logical errors.
P1	Retry metrics are not persisted.	Cannot distinguish baseline demand from network/rate-limit overhead.	Persist logical requests, attempts, retries, 4xx/5xx, timeouts, DNS failures and per-endpoint latency.
P1	Error count semantics are inconsistent.	A roster request can increment <code>errors</code> inside <code>request()</code> and again in the reconcile catch.	Count one terminal error per logical call; track run failure separately.
P1	Public fallback consumer ID is embedded in code.	Easy setup but no central rotation/usage control if DLD changes access terms.	Require the environment variable in production; keep only an explicit local-development path.
P1	No raw schema/lineage layer.	New fields such as <code>ActionUserId</code> and search-match fields can appear unnoticed; cleaned values cannot be reprocessed.	Store request manifest + checksums + sampled/raw payloads with retention and access controls.
P2	Fixed concurrency and broad retry policy.	429s, 403s, DNS and 5xx are retried identically; deep runs may amplify load.	Endpoint-aware backoff with jitter, <code>Retry-After</code> , adaptive concurrency and a request budget.
P2	Project source is date-window constrained.	Absence from the current union cannot be interpreted as delisting, and the official page points prior-year users to Dubai Pulse.	Keep source year/date matches visible, never deactivate on absence, and add licensed historical backfill if complete history is required.
P2	Contact data is personal data.	Saved samples, CRM exports and client products can create privacy/marketing-law exposure.	Purpose limitation, role-based access, audit logging, retention/deletion policy, suppression lists and UAE legal review before outreach/resale.

Specific data-quality observations

- The bulk gateway reported more rows than the crawler could deduplicate: the successful run captured 10,132 of 10,163 reported offices and 33,977 of 34,022 reported brokers before roster reconciliation.
- The current mirror shows 38 zero-broker offices, 9,121 offices with 1-5 active brokers, 835 with 6-20, 175 with 21-100, and 27 above 100; the maximum is 468.

- Broker card issue date is the current term and moves on renewal. Historical “new broker” analysis before the observation baseline is not recoverable from this source alone.
- Office issue date is treated as the original licence date and is the stronger historical lead signal.
- Live response schema includes fields absent from the static interface, demonstrating the need for schema-drift monitoring.
- The project sync stored all 815 unique rows with source hashes and raw JSON; a second identical run should report zero changes, providing a clean idempotency check.

9. Prioritized roadmap and success measures

0-30 days: make the mirror trustworthy

1. Introduce run-scoped raw/staging storage and atomic promotion; prevent partial office/broker snapshots.
2. Record source timestamp, reported total, unique total, logical requests, attempts, retries, status class, latency and failure category per endpoint/stage.
3. Validate `Errors` and `ExceptionId`; add schema-contract tests that tolerate additive fields but alert on removals/type changes.
4. Add scheduler alerts for failure, age-of-last-success and unexpected request growth.
5. Use the supplied Postman collection as the smoke-test baseline. Keep project diagnostics at `P_TAKE=1`; reserve `P_TAKE=-1` for the scheduled four-call snapshot.

30-90 days: productize existing registry intelligence

1. Launch new-brokerage, expiry and roster-change alerts with explicit freshness labels.
2. Validate websites/socials with compliant enrichment and distinguish “missing at DLD” from “not found after validation.”
3. Feed CRM outcomes back into scoring: contact rate, reply, meeting, win, deal value and false-positive reason.
4. Publish account-level timelines: licence, contact/rank/digital changes, broker joins/leaves and outreach history.
5. Add customer-facing data provenance and confidence badges.

90-180 days: expand the data moat

1. Ingest DLD public transaction, rent and developer datasets under confirmed terms; join them to the now-live project dimension.
2. Evaluate the supported Dubai Brokers API against the public gateway on SLA, completeness, terms and total cost.
3. For brokerage software customers, evaluate Rental Index and Trakheesi listing validation/delisting subscriptions.
4. Create market benchmarks: sales/rent price per sq.m., transaction velocity, yields, off-plan share, project pipeline and brokerage growth.

Recommended KPIs

Reliability	Data quality	Commercial
Last successful snapshot age; success rate; p95 stage latency; actual attempts/logical call; DNS/timeout/429 rate.	Unique/reported coverage; duplicate rate; schema-drift events; contact validation rate; stale-record rate; roster completeness.	Lead-to-contact, reply, meeting and win rates by signal; cost per qualified lead; alert engagement; client retention; revenue per data product.

10. Conclusions

1. **The integration now has two very different cost profiles.** Projects are a predictable four-call snapshot; office/broker roster reconciliation remains hundreds of daily calls and more than ten thousand on a deep run.
2. **The registry is commercially useful now.** It can power timely B2B prospecting, compliance, roster intelligence and digital-readiness products without waiting for transaction data.
3. **Freshness and consistency are the immediate constraint.** The most recent scheduled failure produced a partial mirror and misleading run counters. Fix the ingestion contract before selling “real-time” intelligence.
4. **Project intelligence is now a working product surface.** The 815-row current-window union supports developer, launch, supply, progress, area and campaign analysis with explicit source coverage.
5. **The largest next opportunity is joining entities and projects to market outcomes.** Transactions, rents, units/buildings and developer licensing would explain not only who operates and what is being built, but where demand and realized pricing justify action.
6. **Supported DLD APIs matter for production.** The current gateway is effective but behaves like an undocumented public feed. A supported Dubai Brokers subscription should be evaluated for contractual access, stability and SLA.
7. **Personal-data governance must be designed into the product.** Direct broker/contact data can create value, but outreach, resale, retention and saved examples require purpose limitation and legal review.

Appendix A. Field-level extraction dictionary

Office response mapping

Gateway field	Database field / handling	Extracted?
RealEstateNumber	Office.realEstateNumber (primary key)	Yes
LicenseNumber	licenseNumber , cleaned	Yes
OfficeNameEn / OfficeNameAr	nameEn / nameAr	Yes
OfficeIssueDate / OfficeExpiryDate	issueDate / expiryDate , parsed	Yes
OfficePhone / OfficeFax	phone normalized; fax cleaned	Yes
OfficeWebSite	website , URL normalized	Yes
ContactPersonNameEn / Ar	contactNameEn / Ar	Yes
ContactPersonMobile / Email	contactMobile / contactEmail , normalized/validated	Yes
OfficeLogo	logo	Yes
OfficeRankId / OfficeRank	rankId / rank	Yes
AwardsCount	awardsCount , null becomes 0	Yes
YouTubeUrl / FaceBookUrl / InstagramUrl / TwitterUrl / LinkedInUrl	Corresponding URL columns, normalized	Yes
WhatsAppBusinessNumber	whatsapp , phone normalized	Yes
Activities[]	JSON {id,nameEn,nameAr}	Yes
RowIndex / row TotalRowCount	Discarded	No

Broker response mapping

Gateway field	Database field / handling	Extracted?
CardNumber	Broker.cardNumber (primary key)	Yes
CardHolderNameEn / Ar	nameEn / nameAr	Yes
CardIssueDate / CardExpiryDate	cardIssueDate / cardExpiryDate , parsed	Yes
CardHolderPhone / Mobile / Email	phone / mobile / email , normalized/validated	Yes
RealEstateNumber / LicenseNumber	realEstateNumber / licenseNumber	Yes
OfficeNameEn / Ar	officeNameEn / officeNameAr	Yes
OfficeIssueDate / ExpiryDate	officeIssueDate / officeExpiryDate	Yes
CardHolderPhoto	photo	Yes
OfficeRankId / OfficeRank	officeRankId / officeRank	Yes
CardRankId / CardRank	cardRankId / cardRank	Yes
AwardsCount	awardsCount , null becomes 0	Yes
OfficeLogo	Discarded by broker normalizer	No
SearchMatchTypeNameAr / En	Undeclared/discarded	No

RowIndex / row TotalRowCount	Discarded	No
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Project response mapping

Gateway field(s)	Database field / handling	Extracted?
PROJECT_NUMBER	Project.projectNumber primary key	Yes
PROJECT_EN / PROJECT_AR	nameEn / nameAr	Yes
DEVELOPER_NUMBER / DEVELOPER_EN / DEVELOPER_AR	Developer ID and bilingual names	Yes
START_DATE / END_DATE / ADOPTION_DATE / INSPECTION_DATE / COMPLETION_DATE	Five UTC-anchored timestamp columns	Yes
PRJ_TYPE_EN / PRJ_TYPE_AR	Bilingual project type	Yes
PROJECT_VALUE / ESCROW_ACCOUNT_NUMBER	Decimal AED value and string escrow account	Yes
PROJECT_STATUS / PERCENT_COMPLETED	Status and decimal completion percentage	Yes
DESCRIPTION_EN / DESCRIPTION_AR	Bilingual long description	Yes
AREA_EN / AREA_AR / ZONE_EN / ZONE_AR	Bilingual area and authority	Yes
CNT_LAND / CNT_BUILDING / CNT_VILLA / CNT_UNIT / CNT_TOTAL	Five integer inventory counters	Yes
MASTER_PROJECT_EN / MASTER_PROJECT_AR	Bilingual master project	Yes
RN / TOTAL / DEFAULT_SORT	Preserved in rawSource , not duplicated as core columns	Raw only

Appendix B. Postman collection usage

1. Import `HLabs_DXB_DLD_API.postman_collection.json` into Postman.
2. Review collection variables: `baseUrl`, `consumerId`, `pageSize`, `pageIndex`, `officeNumber`, `cardNumber`, `projectFromDate`, `projectToDate` and `projectTake`.
3. Start with page sizes of 2-5 for diagnostics. Do not run full collection iterations with `pageSize=1000` across hundreds of indexes.
4. The project folder parameterizes all four runtime POST variants through `projectDateType`. Use `projectTake=1` for inspection; set it to `-1` only when intentionally retrieving a whole selector window.
5. Each request includes basic envelope and logical-error tests.
6. Each registry request includes a sanitized live-derived response; each project variant includes a public project-data saved 200 response so every returned field and data type can be inspected without rerunning the API.
7. The “point lookup” folder is labelled dormant because those helpers are implemented but unused by the application today.

Appendix C. Sources and evidence

- Repository source: `lib/dld/client.ts`, `lib/dld/sync.ts`, `lib/dld/projects-client.ts`, `lib/dld/projects-sync.ts`, `lib/dld/normalize.ts`, `lib/dld/derive.ts`, `prisma/schema.prisma`, `scripts/install-schedule.ps1`,

and README.md .

- Database: SyncRun , ProjectSyncRun , Office , Broker and Project counts/coverage on 28 July 2026.
- Live gateway probes: /offices/ , /brokers/ and POST /open-data/projects , including four full-year date-selector calls.
- DLD Real Estate Data: <https://dubailand.gov.ae/en/open-data/real-estate-data/>
- DLD API Gateway catalogue: <https://dubailand.gov.ae/en/eservices/api-gateway/>
- DLD API Gateway registration/terms example: <https://dubailand.gov.ae/en/eservices/api-gateway-registration?apId=3>

Audit note: Counts are a point-in-time technical assessment, not a contractual statement by DLD. Live totals and gateway behavior can change. Commercial use of personal/contact data and subscription APIs should be reviewed against current DLD terms and applicable UAE privacy/marketing requirements.